

The Connected Hangar: Ubiquitous Computing and Aircraft Maintenance

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ABSTRACT

In the aviation space, predictive maintenance (PMx) is a strong driver in the push for Internet of Things (IoT) device management systems, artificial intelligence (AI)/machine learning (ML) research, and cloud infrastructure. The potential for this approach to reduce downtime, maximize component lifetime, reduce man-hours on diagnosis and repair, and optimize supply chains and scheduling has driven massive investments across the industry. And yet, the challenges in delivering on these promises with the available data and technology should also not be minimized. To reach its full potential, PMx implementers must understand what predictions can be derived from the available data, what maintenance actions may be driven by those predictions, and how the predictions should be presented to the appropriate decision makers in ground operations and the logistics chain.

In this paper, we will examine the current state of data within the aviation PMx space, variations in component level coverage, and how that translates to the type, volume, and timeliness of data and computational infrastructure necessary to provide *right time* predictions and analytics to maintainers, supply chain managers, and operators. We will also examine the specific challenges for PMx in the aviation space with respect to data availability, equipment variability, use variability, and maintenance action coding that can affect the ability of operators to derive value from a PMx program.

CCS CONCEPTS

• **Applied computing** → **Physical sciences and engineering**: Avionics;

KEYWORDS

Aircraft maintenance, aviation, human factors

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1 INTRODUCTION

To maximize equipment reliability, maintenance programs have long utilized data to predict and identify failures, estimate useful life, intelligently order parts, and select the correct timing for maintenance activities. In aviation, maintainers also rely on a combination of line checks, time- and usage-based maintenance strategies, and pilot reports to determine the appropriate actions to maximize time-on-wing (TOW). Modern computing capacity and onboard sensor capabilities permit anomaly detection and crew awareness of issues both in-flight and upon landing. The current capability to collect machine data from aviation equipment is vast with an engine capable of producing terabytes of data per flight hour. At the same time, many other types of data, both legacy and modern, exist in various forms across the aviation maintenance domain, and the extent to which data is utilized is inconsistent across the industry. The path of effective maintenance, as described in [4], suggests an alignment of "increasingly efficient maintenance effort, realistic resources, and doing the right thing." We will examine the current state of available data resources and how they are being used in the effort to do the right thing at the right time.

2 MAINTENANCE STRATEGIES AND THE AVIATION MAINTENANCE CHECK SYSTEM

Aircraft maintenance checks are the standard inspections that every aircraft receives to be cleared to fly. In addition to the pre- and post-flight checks, also known as line checks, there are checks that take place at regular calendar or usage intervals known as *ABCD checks*. These checks are typically defined by the manufacturer initially but can be modified by the operator in a process with the governing regulatory body. A-checks are the most frequent and can take an aircraft out of service for several hours while D-checks are a full refit of the aircraft that may take an aircraft out of service for

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weeks. The scheduling of these maintenance activities requires the coordination of maintainers, aircraft, flight crews, hangars, and supply chains. This is typically a highly manual task pulling in representatives across all of the aforementioned groups and is generally not optimized. [2]

Preventative maintenance (PM) tasks such as these are now also supplemented with condition-based maintenance (CBM) and predictive maintenance (PMx), which use sensor devices on subsystems of the aircraft to determine maintenance requirements or to give an estimate of remaining useful life (RUL) of a component. [1] Predictive maintenance strategies can also incorporate pieces of data outside the component in question such as data from maintenance logs, information from flight acquisition units, and fleet-wide analysis to determine if an action needs to be taken by the maintainers. Ideally, the computing required to do these tasks would be handled during the flight or in the time it takes to prepare the aircraft for the next flight to maximize on-time departures and minimize downtime.

3 DATA AVAILABILITY

Today, many types of ubiquitous computing technologies have been incorporated into aviation maintenance. Sensor networks, mobile "smart" devices, and ERP systems were outlined as highly suited for aircraft maintenance by [3], and are now commonly seen across many commercial operators. This newer, highly structured data can exist alongside data available from older methods of reporting such as manually-entered sensor observations as well as legacy data such as handwritten maintenance logs and inspection reports. A comprehensive approach to the data must take into account availability as well as the quality, quantity, and utility when driving a maintenance action.

The figure below gives an overview of the data flowing around aircraft maintenance from onboard sensors, maintenance logbooks, inspection drones, "smart" toolboxes, imagery from structural inspection reports, handheld Bluetooth-enabled sensors that commercial carriers are beginning to adopt, and the ERP systems for managing part supply chains. Much of this data could be used to augment a maintenance program and has been to varying degrees. In addition, data systems in the aviation space typically accommodate vast quantities of historical information from past inspections, but it can be challenging to get it out in a useful format from legacy systems and often requires extensive conditioning to be usable.

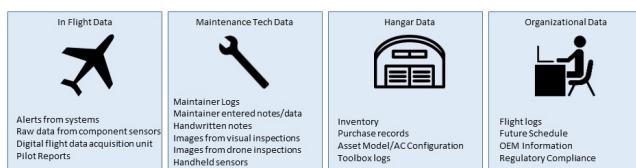


Figure 1: Type and Nature of Aviation Maintenance Data.

However, there are still many situations in which potentially available and useful data are not collected in a usable way or is not appropriately connected to other data sources. One example is data from handheld sensors such as tire pressure gauges. Typically, data taken from these sensors are not consistently entered into any system of record as it is often meant to be a very quick check,

and it is left to the individual maintainers to decide if a reading is necessary and relevant to record as is shown in Table 1. There are efforts to convert some of those sensors into smart sensors that can automatically enter readings into a database, but currently, tire pressures in commercial aviation are using the more traditional approach, and tire pressure is often not recorded.

The differences in data type and collection result in different kinds of challenges between components all the way to fleet level. Certain components, such as the power plant, can be highly sensed and capable of producing many terabytes of data per flight hour. This data is often edge processed with only event data captured and recorded. Other components in the system may be generating large quantities of data. However, this data may not be captured or used in any record or, even if fully captured, may never be analyzed. This could be viewed as a potential investigation area for data analysis or as a waste of system resources. Many components in the aircraft such as structural components or consumable components such as tires may not generate any data from sensors but are monitored through regular inspection. The inspection reports may be stored as a structured form, a scanned report, or, in some cases, a paper form stored in the hangar. The onboard Flight Data Acquisition Unit (FDAU) is required by regulation to capture certain aspects of aircraft flight such as time, position, heading, attitude, and critical control parameters. These are required by law to only be kept for a limited amount of time; the retention of these artifacts after that time period is not guaranteed to be consistent.

The maintenance and hangar support teams generate many data streams on their own. The hangar will have orders and invoices for parts and consumables. Maintenance teams will keep time sheets, logs, inspection reports, and compliance documents and reports, some of which only exist in paper form. Some operators have also begun using smart tool and tool management systems that send data and tool usage information straight to a database.

Inadequate job coding can also hinder a predictive maintenance program. Historical maintenance actions can be used for a variety of purposes in both condition-based and predictive maintenance, but in many cases, the classification of such actions can be inconsistent or contain information inadequate to assess condition. One example is that of tire changes. In investigations across multiple maintenance programs we have observed that the position of the tire change is not always recorded, making it impossible to determine the RUL for any individual tire as well as preventing a calculation of mean time between failure (MTBF) in fleet-wide statistics. Many RUL prediction techniques are limited to applications with fixed operating conditions, and adequate underlying data structure to capture those conditions may not be present to train a machine learning model; usefully, there are some techniques that can be used to overcome these deficiencies. [5]

Within a particular airframe under a single operator, a uniform fleet becomes heterogeneous due to retrofits and upgrades applied at different times due to differences in usage, wear, or damage. This can complicate efforts to baseline and characterize fleet health due to the variation within the population. Heterogeneity can also cause unforeseen complications in data as replacement components may add new or changes to existing data streams that may not be reflected in the system of record. The system of record may not also be updated to take advantage of sensor data that comes online

Table 1: Example Analysis of Tire Data availability for maintenance

Stream	Type of Data	Data Feed	Completeness	Reliability
Handheld Sensors	PSI, Temp, Tread Depth	Human Recorded	Inconsistent	Low Reliability
On board Sensors	Automated warnings for low pressure	Automatic	Available in Database	Reliable
Configuration	Serial Number, Position, Hours used, Number of Cycles	Human Recorded	Available in Database	Semi Reliable
Asset/Inventory	Number in Stock, Location, Usage, hangar purchase logs	Software processed and recertified during inspection and inventory	Available in Database	Semi Reliable
Operational Data	Pilot assignment, technician servicing tire	Automated/Sensor	Available in Database	Reliable
Schedule	airport locations and dates	Automatic	Available in Database	Reliable
Maintenance History	Maintenance shop data, installation dates, maintenance logs, hangar purchase logs	Human Recorded	Available in Database Position of tire change not consistently recorded	Semi reliable

as newer components replace unsensored legacy components. This misses a monitoring opportunity that could remain overlooked for years.

4 CONCLUSION

Having a realistic characterization of the kinds and nature of data available in the aviation maintenance ecosystem is useful to those desiring to develop advanced maintenance programs that rely on extensive amounts of clean and structured data.

Aircraft maintainers are incorporating ubiquitous computing technologies into their operations in a drive towards less reactive and more predictive maintenance strategies. While there have been many advances in the collection of data in some highly sensed subsystems such as the engine, subsystems that rely on visual inspection and handheld sensors may not have as much coverage, which can limit options for developing an optimized maintenance strategy. Additionally, organizational limitations such as how maintenance actions are recorded can hinder the calculation of key statistics necessary for a predictive maintenance program. It requires the combined effort of maintainers, subject matter experts, and data

scientists and engineers to identify and address these shortcomings in order to maximize the utility of the available data.

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